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ECS 132

This test has 0-6 problems, the last page has problem 6(don't forget). You will use **your y** variable which is calculated by taking just the **last digit** of your student **id** 9 **modding it by 7** and then **adding 6** to it. Everything in the test is self-explanatory. If you are confused, state your assumption and answer the question. We will grade with your assumption in mind. **We cannot answer any questions related to your interpretation.**

What is your y ? 8

Do not forget to fill in your y in the problems below. **Do not solve with y being a variable. Fill in your constant value.**

***Show all work and calculations, if it doesn't explicitly say to simplify you do not need to simplify.**

Problem 0 (15 pts)

What is your y ? 8

a.) Assume a website allows passwords made of lowercase letters and digits only. A password is considered **invalid** if it either ends with two 0s or begins with two 1s. You randomly generate a password of length y+3, what ~~is the pro~~bability that it is a valid password?

25 letters   9 digits   11 values

1st = 34

$$\left(1 - \frac{32(9 + 32(9)}{34^{11}}\right.$$

b.) What is the probability that, when rolling two $\overset{2\ 8}{\text{y-sided}}$ dice, the total number of dots is greater than or equal to 4?

1 1

2 1

1 2

Answer = $1 - \frac{3}{64}$

Problem 1 (30 pts)

What is your y ? 8

a.) You are planning a conference and have 2 meeting rooms. Such that the first room fits $\overset{24}{3y}$ people, the second fits $\overset{36}{4y+4}$. The CEO's of Facebook and Google are both attending your conference, what is the probability that they are not placed in the same conference room.

$1 - (58C22)(7(C36) + (60C24)(34C34)$ Total = 60

What is your y ? 8

b.) You notice that a bus always has y+7,2y+3, or 3y people riding it when it arrives at your stop. Let X=number of people that arrive at your bus stop. Given P(X=y)=0.2 P(X=2y)=0.3 P(X=3y)= 0.5
Find Var[X] using **this equation ONLY** Var[x]=E[(x-EX)^2]. DO NOT SIMPLIFY.

15 19 24

+7 +8

$E[X] = 15(0.2) + 19(0.3) + 24(0.5) = Ex$

$Var[X] = (15 - Ex)^2 0.2 + (19 - Ex)^2 0.3 + (24 - Ex)^2 0.5$

c.) Suppose we deal a 5-card hand from a regular 52-card deck. What is P(exactly 3 hearts or exactly 2 kings) $= P(3h) + P(2k) - P(1k \cap 2h) - P(3h \cap 2k)$

$P(3h) = 13C3 + 39C2$

$P(2k) = 4C2 + 48C3$

$P(1k \cap 2h) = 4C1 + 12C2 + 36C2$

$P(3h \cap 2k = 13C3 + 4C2$

for i in reps

Provide code to check your analytical result via simulation.

```
for (i in 1:nreps)
  x = sample(1:52)
  hand[]
  if hand ∈! x
    hand = x

  for h in hand
   if(h%13==0)
      kings = king+1
   if h ≥13
      hearts = hearts +1
  if (hearts == 3 || kings = 2 && || hearts==3 && kings==2
    value = value+1
```

Problem 2 (30 pts)

What is your y ? 8

You are rolling a fair y+3-sided dice in a casino where you will win $100 if you **roll 3 or 4.** 11

a.) What is the probability that you will **win on the** 7th and again on the 8th roll (regardless of other wins or losses)?

$\frac{2}{11} \cdot \frac{2}{11} = \frac{4}{121}$

b.) What is the expected value of the number of rolls it will take you to win?

$\frac{2}{11}$ games your expected to win

c.) What is the probability you will win for the first time on the 8th roll?

not win first 7 games $\underbrace{\left(1 - \frac{2}{11}\right)^7}$ $\frac{2}{11}$ win on the 8th

d.) What is the expected value of the amount of money you will win if you play 4 games? $100\left(\frac{2}{11}\right) + 100\left(\frac{2}{11}\right) + 100\left(\frac{2}{11}\right) + 100\left(\frac{2}{11}\right)$

e) What is the variance value of the amount of money you will win if you play 4 games? $E[X] = \frac{200}{11}$ $E[X] = 100^2 \frac{2}{11}$

$\boxed{4[V[X]] = 4\left(100^2\left(\frac{2}{11}\right) - \left(\frac{200}{11}\right)^2\right)}$

f.) Let X=the amount of money you win on a single game. What is Var[2X+**y**] = $Var[2X]$

$E[X]$

$4 Var[X] =$

Problem 3. (20 pts) What is your y? 8

You buy iHealth covid test and test negative ~~before~~ today's midterm. given iHeath reported that in people sick with covid - 8% tested negative, while people who are covid free -tested positive (7+y)% 15 of the time. Currently, in the population, **(8+y)**% 16 of people have covid. . What is the probability that you are currently **infected** with covid?

Don't just provide an answer state your probabilities show how you calculate the final solution.

$P(n|\bar{I}) = 1-0.15$

$P(\text{infected} \mid \text{neg}) \rightarrow P(I|N) =$

$P(\bar{I}) = 1 - P(I)$

$= 1 - 0.16$

$$\frac{P(N|I)\,P[I]}{P(N|I)\,P I + P(n|\bar{I})\,P(\bar{I})} = \frac{0.08\,(0.16)}{0.08\,(0.16) + 0.85\,(84)}$$

Problem 4.) (20 pts) What is your y? 8

You are playing a game at a carnival, the game involves rolling a 3 sided dice where: Rolling a

2 has probability of $\frac{1}{2y}$ $\frac{1}{16}$

3 has probability of $\frac{10}{16}$ <FILL IN THE ANSWER>

6 has probability of $\frac{y-3}{2y}$ $\frac{5}{16}$

IF your dice face is a 3 you will receive and flip a fair coin else you will flip a coin where Probability(Heads)=(33+y)%. Let X be a random variable that represents the number of dots you see on a face of the dice, and let Z be the indicator on tails(1 if tails).

$1 = T$

$0 = H$

a.) What is E[X]

$2\left(\frac{1}{16}\right) + 3\left(\frac{10}{16}\right) + 6\left(\frac{5}{16}\right) = E(x)$

b.) What is E[XZ]

$2 < \begin{matrix} 1 & \frac{1}{2} \\ 0 & \end{matrix}$ $\frac{41}{100}$

$3 < \begin{matrix} 1 \\ 0 \end{matrix}$

$6 < \begin{matrix} 1 & \frac{41}{100} \\ 0 & \end{matrix}$

$\left(2\left(\frac{1}{16}\right)\frac{1}{2} + 3\left(\frac{10}{16}\right)\frac{41}{100} + 6\left(\frac{5}{16}\right)\frac{41}{100} = E(XZ)\right.$

c.) Find Cov(X,Z)

$E(2) = \frac{1}{16}\left(\frac{1}{2}\right) + \frac{10}{16}\left(\frac{41}{100}\right) + \frac{5}{16}\left(\frac{41}{10}\right)$

$E(XZ) - E(X)E(Z)$

used definition and X is dependent on Z cause X changes Z probability (1) to happen.

Problem 5(15): 16

a.) Assume you are rolling a fair **2y** sided dice. Let X be the number of dots on the dice. If you get a 2 you will receive a fair coin; otherwise, you will receive an unfair coin with P(heads)=(y+30)%. Let Z be a variable that equals 10 if the outcome of the coin is heads and 5 if it is tails. **Simulate P(Z+X>13 or X>5) ONLY do not solve analytically.**

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c==1 is heads
c==2 is tails

```
Declaration
num=0
c=0
x=0
z=0
for(i in 1:nreps)
X = Sample(1:6)
if(x==2)
  c = Sample(1:2)
  if c==1 z=10 else z=5
else
  c = Sample(1:100)
  if c <= 38
    z=10
  else
    z=5
if(z+x>13 || x>5)
  num = num+1
```

Problem 6.)(35 pts) What is your y? 8

a) For the Aloha question, assume that there are 2 nodes. Assume that one node is inactive at start of epoch 4. What is P(X4=1)? You may use p and q without substituting for actual values.

P(X4=1)

n_1	n_2			
X	✉	s 0	$(1-p)(1-q)(q) = v$	
NA		n 1	$(1-p)(1-q)^2 = s$	
AN		s 1	$p(1-q)q = h$	
AS		n 2	$p(1-q)^2 = x$	
		s 2	$pq^2 = z$	
		n 1	$(pq)(1-q) = u$	

$$P(X4=1) = \frac{s+h+u}{s+h+u+z+x}$$

b.) Now assume you have only 2 nodes and both are inactive (i.e. X0=0). Simulate to find the Prob(C=1|X2!=X1) where C is coalition count. You may augment the code below.

$P(C=1 \mid X_1 \neq X_2)$

```
# finds P(X1 = 2), P(X2 = 2) and P(X2 = 2|X1 = 1) in ALOHA example
sim <- function(p,q,nreps) {
   countx2eq2 <- 0
   countx1eq1 <- 0
   countx1eq2 <- 0
   countx2eq2givx1eq1 <- 0
   # simulate nreps repetitions of the experiment
   for (i in 1:nreps) {
      numsend <- 0  # no messages sent so far
      # simulate A and B's decision on whether to send in epoch 1
      for (j in 1:2)
         if (runif(1) < p) numsend <- numsend + 1
      if (numsend == 1)  X1 <- 1
      else X1 <- 2
      if (X1 == 2) countx1eq2 <- countx1eq2 + 1
      # now simulate epoch 2
      # if X1 = 1 then one node may generate a new message
      numactive <- X1
      if (X1 == 1 && runif(1) < q) numactive <- numactive + 1
      # send?
      if (numactive == 1)
         if (runif(1) < p) X2 <- 0
         else X2 <- 1
      else {  # numactive = 2
         numsend <- 0
         for (i in 1:2)
            if (runif(1) < p) numsend <- numsend + 1
         if (numsend == 1) X2 <- 1
         else X2 <- 2
      }
      if (X2 == 2) countx2eq2 <- countx2eq2 + 1
      if (X1 == 1) {  # do tally for the cond. prob.
         countx1eq1 <- countx1eq1 + 1
         if (X2 == 2) countx2eq2givx1eq1 <- countx2eq2givx1eq1 + 1
      }
   }
   # print results
   cat("P(X1 = 2):",countx1eq2/nreps,"\n")
   cat("P(X2 = 2):",countx2eq2/nreps,"\n")
   cat("P(X2 = 2 | X1 = 1):",countx2eq2givx1eq1/countx1eq1,"\n")
}
```

if(runif(1) < a) numactive ← numactive +1

if numactive = 1

(runif(1) < p) numsend ← numsend +1

if numactive = 2

for j in 1:2

(runif(1) < a) numsend ← numsend +1

If numsend == 2

collision = 1

if numactive == 1

if runif(1) < q numsend = numsend +1

Numactive = numactive -1.

if (X1 != X2 && numsend == 2

collision = collision +1

If col

else X1 != X2 && collision = 1

Value = Value +1

Picking

Picking with replacement $\prod_{i=1}^{n} X_i$
Picking without replacement where order matters $nPr = \frac{n!}{(n-r)!}$
Picking without replacement where order does not matter $nCr = \binom{n}{r} = \frac{n!}{(n-r)!r!}$

Set Theory

Let A be an event, and S be the sample space
Complement
$|A| = |S - A|$
Union$|A \text{ or } B| = |A \cup B| = |A| + |B| - |A \cap B|$

Difference A - B = All elements in A minus/remove elements in B
Complement of sample space $S = \varnothing$

Probability

Naive def of prob: $P(A) = \frac{|A|}{|B|}$
$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$
P(A and B) $= P(A \cap B) = P(A \mid B)P(B)$
$P(A \cap B) = P(B \mid A)P(A)$
if A & B are indepedent $P(A \cap B) = P(B)P(A)$

if A&B are disjoint $P(A or B) = P(A \cup B) = P(A) + P(B)$
for any A, B $P(A or B) = P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Empty set probability Given that $E = \varnothing$, $P(E) = 0$
complement $P(A) = 1 - P(A)$

$$E(X) = \lim_{n \to \infty} \frac{X_1 + X_2 + X_3 + \ldots + X_n}{n}$$
$$= \sum_{c \in A} c \cdot P(X = c)$$
$$E(U + V) = E(U) + E(V)$$
$$E(aU) = a \cdot E(U)$$
$$E(b) = b$$
$$E(aX + B) = a \cdot E(X) + b$$

given U and V indep. then $E(UV) = E(U) \cdot E(V)$

$$E(g(X)) = \sum_{c \in A} g(c) \cdot P(X = c)$$

$$\text{Var}(X) = E(X^2) - (E(X))^2$$
$$\text{Var}(cX) = c^2 \text{Var}(X)$$
$$\text{Var}(X + a) = \text{Var}(X)$$

Covariance
$Cov(X, Y) = E((X - E(X))(Y - E(Y)))$
$= E(XY) - E(X)E(Y)$

$$P_X(k) = P(X = k)$$

Indicator Variable Family

$$P_X(k) = \begin{cases} p & \text{if } k = 1, \\ 1 - p & \text{if } k = 0, \\ 0 & \text{otherwise.} \end{cases}$$
$$E(X) = p$$
$$\text{Var}(X) = p(1 - p)$$

Geometric Variable Family

$$P_X(k) = (1 - p)^{k-1} p$$
$$E(X) = \frac{1}{p}$$
$$\text{Var}(X) = \frac{1 - p}{p^2}$$

Uniform Discrete Variable Family

$$P_X(k) = \begin{cases} \frac{1}{b-a+1} & \text{if } k \in [a, a+1, a+2, \ldots, b-1, b], \\ 0 & \text{otherwise.} \end{cases}$$
$$E(X) = \frac{a + b}{2}$$
$$\text{Var}(X) = \frac{(b - a)(b - a + 2)}{12}$$

Poisson Variable Family

$$P_X(k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k \in \{0, 1, 2, 3, \ldots\}$$
$$E(X) = \lambda$$
$$\text{Var}(X) = \lambda$$

Binomial Variable Family

$$P_X(k) = \binom{n}{k} p^k (1 - p)^{n-k}$$
$$E(X) = np$$
$$\text{Var}(X) = np(1 - p)$$

var[x+y]=var[x]+var[y]+2cov(x,y)

P(class|feature)=
P(class and feature)/ P(feature)